



COURSE DESCRIPTION FORM: [MT1008]: [Multivariable Calculus]

INSTITUTION: FAST School of Computing, National University of Computer and Emerging Sciences, Islamabad Campus

Computer Science (BSCS)-**Spring-2024**

PROGRAM(s) TO BE EVALUATED

Course Description

Course Code	MT-1008	
Course Title	Multivariable Calculus	
Credit Hours	3	
Course Instructors	Dr. Hamda Khan, Dr. Imran Shahzad, Mr. Arif Hussain	
Grading Policy	Absolute	
Policy about missed assessment items in the course	Retake of missed assessment items (other than sessional/ final exam) will not be held. A student who misses an assessment item (other than sessional / final exam) is awarded zero marks in that assessment item i.e., late submission will not be accepted. For missed sessional/ final exam, exam retake/ pre-take application along with necessary evidence are required to be submitted to the department secretary. The examination assessment and retake committee decide the exam retake/ pre-take cases.	
Course Plagiarism Policy	Plagiarism in projects or sessional/ final exam will result in F grade in the course. Plagiarism in an assignment will result in zero marks in the whole assignments category.	
Prerequisites by Course(s) or Topics	Calculus and Analytical Geometry	
Assessment Instruments with Weights (homework, quizzes, sessional exams, final exam, assignments, etc.)	Assessment with the weight.	
	Assessment Type	Weight
	Assignments (4-5)	05
	Quizzes (7-9)	08
	Project	07
	Sessional I	15
	Sessional II	15
	Final Exam	50
Course Coordinator	Dr. Hamda Khan	
URL (if any)	-	

Course Catalog Description	Multivariable functions, Limit and continuity in higher dimensions, Partial derivatives, Extreme values and their applications, Chain rule, Directional Derivatives and Gradient Vectors, Applications of Gradient, Tangent Planes and Normal Lines, Linearization, Constrained Maxima and Minima, Lagrange Multipliers, Double integrals over rectangular and general regions, Area by double integration, Double integral in polar form, Triple integrals, Automatic Differentiation, Computational Graph, Line Integrals, Vector Fields, Gradient Fields, Path Independence, Conservative Fields, Line Integral in Conservative Fields, Potential Functions, Divergence, Parametrizations of Surfaces, Surface Area, Surface Integrals, Surface Integrals of Scalar Functions, Surface Integrals of Vector Fields, The Curl Vector Field, Divergence in Three Dimensions, Divergence Theorem.																								
Textbook(s)	Thomas Calculus, 13th ed., by George B. Thomas Jr, Maurice D. Weir and Joel Hass, Pearson.																								
Reference Material	1. Calculus (Sixth Edition) By Swokowski. 2. Learning Scientific Programming with Python, Christian Hill, University College London.																								
Course Goals	A. Course Learning Outcomes (CLOs) After completion of the course, the student shall be able to: 1. Understand the basic concepts associated with Calculus problems in several variables. 2. Apply appropriate techniques for solving problems in several variables including differentiation, double and triple integrals. 3. Adapt modern programming languages and computing tools to apply appropriate mathematical techniques on real life problems. 4. Organize and prioritize work effectively as an individual or as a member/leader of a technical team. B. Program Learning Outcomes (PLOs) <table><tr><td>PLO 1</td><td>Computing Knowledge</td><td>Apply knowledge of mathematics, natural sciences, computing fundamentals, and a computing specialization to the solution of complex computing problems.</td><td>✓</td></tr><tr><td>PLO 2</td><td>Problem Analysis</td><td>Identify, formulate, research literature, and analyze complex computing problems, reaching substantiated conclusions using first principles of mathematics, natural sciences, and computing sciences.</td><td></td></tr><tr><td>PLO 3</td><td>Design/ Develop Solutions</td><td>Design solutions for complex computing problems and design systems, components, and processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.</td><td>✓</td></tr><tr><td>PLO 4</td><td>Investigation & Experimentation</td><td>Conduct investigation of complex computing problems using research based knowledge and research based methods.</td><td></td></tr><tr><td>PLO 5</td><td>Modern Tool Usage</td><td>Create, select, and apply appropriate techniques, resources and modern computing tools, including prediction and modelling for complex computing problems.</td><td>✓</td></tr></table>					PLO 1	Computing Knowledge	Apply knowledge of mathematics, natural sciences, computing fundamentals, and a computing specialization to the solution of complex computing problems.	✓	PLO 2	Problem Analysis	Identify, formulate, research literature, and analyze complex computing problems, reaching substantiated conclusions using first principles of mathematics, natural sciences, and computing sciences.		PLO 3	Design/ Develop Solutions	Design solutions for complex computing problems and design systems, components, and processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.	✓	PLO 4	Investigation & Experimentation	Conduct investigation of complex computing problems using research based knowledge and research based methods.		PLO 5	Modern Tool Usage	Create, select, and apply appropriate techniques, resources and modern computing tools, including prediction and modelling for complex computing problems.	✓
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	and Normal Lines, Linearization, Constrained Maxima and Minima, Lagrange Multipliers, Applications of Constrained Optimization. Use of modern tools / technology: 3D plot, calculating the partial derivative, gradient, finding extrema of multivariable function.			
	Double integrals over rectangular and general regions, Area by double integration, Double integrals in polar form, Triple integrals, Applications of multiple integrals, Automatic Differentiation. Use of modern tools / technology: calculating the integrals of multivariable functions	4	12	1,2,3,4
	Computational Graph, Line Integrals, Vector Fields, Gradient Fields, Path Independence, Conservative Fields, Line Integral in Conservative Fields, Potential Functions, Divergence, Parametrizations of Surfaces, Surface Area, Surface Integrals, Surface Integrals of Scalar Functions. Use of modern tools / technology: Use of Numpy for surface integrals, Visualize divergence fields with Matplotlib, Compute gradients using NumPy.	3	9	1,2,3,4
	Surface Integrals of Vector Fields, The Curl Vector Field, Divergence in Three Dimensions, Divergence Theorem. Use of modern tools / technology: Use NumPy to compute the curl and divergence of a vector field.	2	6	1,2,3,4
	Total	15	45	
Programming Language for Assignments (if any)	Python (SymPy, Scipy, Numpy)			
Class Time Spent (in percentage)	Theory	Problem Analysis	Solution Design	Social and Ethical Issues
	35	30	30	5
Oral and Written Communications	Every student is required to submit at least 01 written report of typically 10-12 pages.			

Weeks	Contents/Topics	Courseware Events (Quiz/ Assignment/ Project)
Week 01	Function of Several Variables, Domain & Range Open, Closed Regions, Bounded/Unbounded Regions, Graphs, Level Curves, Contours of Functions of Two variables, Ex 14.1.	
Week 02	Limit, Path Test for Limit, Continuity, Ex 14.2, Partial Derivatives, Ex 14.3, Extreme Values and Saddle Point, Derivative Tests for Local Extreme Values.	Quiz # 1
Week 03	Absolute Maxima and Minima on Closed Bounded Region, Ex 14.7, Constrained Maxima and Minima, Method of Lagrange Multipliers, Ex 14.8, Chain Rule.	Assignment # 1 Quiz # 2
Week 04	Ex 14.4, Directional Derivatives and Gradient Vector, Applications of Gradient, Ex 14.5.	Quiz # 3
Week 05	Tangent Planes and Normal Lines, Linearization, Ex 14.6, Automatic Differentiation.	Assignment # 2 Quiz # 4
Week 06	Computational Graphs, Revision of Sessional I topics.	Sessional I
Week 07	Recalling integration of function of one variable, Double Integrals over Rectangular Regions, Ex 15.1, Intro to nonrectangular regions.	Assignment # 3
Week 08	Double Integrals Over Nonrectangular Regions, Ex 15.2, Area by double integration, Ex 15.3.	Quiz # 5
Week 09	Intro to polar coordinates and polar curves, Double Integrals in Polar form, Changing Cartesian Integrals into Polar Integrals, Ex 15.4, Triple Integrals in Rectangular Coordinates, Average value of a function in space.	Assignment # 4 Quiz # 6
Week 10	Ex 15.5, Volume of a Region in space, Applications of multiple integrals.	Quiz # 7
Week 11	Line Integrals, Ex 16.1.	Assignment # 5 Sessional II
Week 12	Vector Fields, Gradient Fields, Line Integrals of Vector Fields, Ex 16.2, Path Independence, Conservative Fields, Line Integral in Conservative Fields.	Project
Week 13	Potential Functions, Ex 16.3, Divergence, Parametrizations of Surfaces, Surface Area, Ex 16.5.	Quiz # 8
Week 14	Surface Integrals, Surface Integrals of Scalar Functions, Surface Integrals of Vector Fields, Ex 16.6.	
Week 15	The Curl Vector Field, Divergence in Three Dimensions, Divergence Theorem, Ex 16.8. Revision.	